

Gonçalo Pereira

Embedded Systems · Network Programming · Backend Development

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Embedded software engineer working in C on telecom network gateways running embedded Linux. Background in systems and network programming from 42 Lisboa, and hands-on experience designing, shipping and operating a production Go service with its own CI, deployment and monitoring.

EXPERIENCE

Embedded Software Engineer · Streambow

Dec 2025 – Present · Sintra, Portugal

- Develop embedded C software for telecom network gateway devices running embedded Linux
- Integrate containerised components with Docker into network diagnostics pipelines
- Debug networking behaviour through low-level system and packet analysis

Student Supervisor · 42 Lisboa

Jul 2024 – Oct 2024 · Lisbon, Portugal

- Reviewed and evaluated C and systems programming projects ensuring specification compliance
- Delivered technical presentations to audiences of 100+ students
- Coordinated staff workflows supporting student progress initiatives

PROJECTS

ft_ping & ft_traceroute · C · Raw Sockets · ICMP · Network Internals

2025

- Reimplemented ping and traceroute using raw sockets and handcrafted ICMP packets
- Implemented TTL control, checksum computation and round-trip timing without libc abstractions

malloc · C · Memory Management · mmap · pthreads

2025

- Thread-safe drop-in replacement for malloc, free, realloc and calloc, built over mmap
- Pre-allocated TINY/SMALL/LARGE zones, mutex-guarded metadata, allocation-map debug tooling

Minishell · C · Process Management · Signals · Parsing

2024

- Built a POSIX-style shell with pipes, redirections, built-ins, environment handling and job signals
- Designed a tokeniser and AST-based execution pipeline driving fork/exec workflows

ft_irc · C++ · TCP Sockets · poll() · Protocol Design

2024

- Built an IRCv3 server handling many concurrent clients from a single poll() loop over non-blocking sockets
- Implemented channels, operator modes and command parsing; validated against real clients

dune · C++ · Systems Architecture · State Machines

2026

- Designing a 45k-line C++ engine for a complex turn-based board game
- Phase state machine and event bus, with interaction adapters for TTY, scripted and recorded play

Quest Board · [dnd.fontao.net](#) · Go · PostgreSQL · React · OpenAPI · Docker

2026

- Designed, shipped and operate a production web app: a React SPA compiled into a single Go binary
- Contract-first API — one OpenAPI spec generates the Go server interface and the typed TS client
- OAuth2, local accounts and TOTP 2FA; CI-gated merges, staging and production deploy pipelines

Homelab Observability · Prometheus · Grafana · Docker Compose

2026

- Monitoring stack for the above: application, host and per-container metrics, 30-day retention
- Alert rules provisioned as code; Grafana behind Cloudflare Access, Prometheus never public

EDUCATION

42 Lisboa · Software Development · Systems & Network Programming

Oct 2023 – May 2025 · Lisbon, Portugal

- Project-based curriculum covering C/C++, operating systems, networking and algorithms
- Specialisations in Linux internals and network protocols, and in cybersecurity and reverse engineering

SKILLS

Languages C, C++, Go, Python, Rust, JavaScript/TypeScript, x86-64 Assembly, Bash, SQL

Systems Embedded Linux, Linux internals, syscalls, signals, process and memory management

Networking Raw sockets, TCP/IP, ICMP, protocol implementation, packet analysis (Wireshark, tcpdump)

Backend Go (chi, sqlc, pgx), PostgreSQL, REST and OpenAPI, Django, WebSockets, OAuth2/JWT/TOTP

Infrastructure Docker, Kubernetes (K3s), Prometheus, Grafana, GitHub Actions CI/CD, NGINX

Tooling Git, GDB, Valgrind, Make, contract-driven code generation

Portuguese (Native) · English (Fluent)